SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 25%

Duration: 30 Minutes

Questions: 20 Total Marks: 20

Annexure A

MCQ Test

Code of Conduct:

I KISHORE KUMAR V(192521371) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Kishore V

Signature of the student:

Artificial Intelligence (CSA17) – CO2 MCQ Test: Answer Key

1. In informed search strategies, some algorithms use only the heuristic value to decide which node to expand next. Which algorithm follows this principle?

A) Breadth First Search
B) Uniform Cost Search
C) Greedy Best-First Search
D) Depth First Search

Answer: C) Greedy Best-First Search

2. What is the evaluation function used in A* search?

A) f(n) = g(n)
B) f(n) = h(n)
C) f(n) = g(n) + h(n)
D) f(n) = g(n) × h(n)

Answer: C) f(n) = g(n) + h(n)

3. What property ensures optimality in A* search?

A) Completeness
B) Admissibility
C) Depth limit
D) Backtracking

Answer: B) Admissibility

4. Which of the following is a local search algorithm?

A) A* Search
B) Hill Climbing
C) Breadth First Search

D) Uniform Cost Search

Answer: B) Hill Climbing

5. Which algorithm is commonly used for decision-making in adversarial search?

A) Greedy Search
B) Minimax Algorithm
C) Backtracking
D) Uniform Search

Answer: B) Minimax Algorithm

6. A delivery robot selects paths based only on the estimated distance to the destination without considering obstacles already encountered. Which algorithm is the robot using?

A) A* Search
B) Greedy Best-First Search
C) Uniform Cost Search
D) Depth Limited Search

Answer: B) Greedy Best-First Search

7. A GPS navigation system combines the distance already travelled and an estimate of the remaining distance. Which algorithm best represents this approach?

A) Depth First Search
B) A* Search
C) Hill Climbing
D) Backtracking

Answer: B) A* Search

8. A university is scheduling exams with constraints such as no overlapping exams and limited rooms. This problem is best modeled as:

A) Game Playing Problem
B) Constraint Satisfaction Problem
C) Heuristic Search Problem
D) Optimization Problem

Answer: B) Constraint Satisfaction Problem

9. An AI-based chess program evaluates all possible moves and anticipates the opponent's responses, assigning utility values to game states. Which algorithm is being applied?

A) Greedy Search
B) Hill Climbing
C) Minimax Algorithm
D) Breadth First Search

Answer: C) Minimax Algorithm

10. A robot explores an unknown environment and updates its knowledge as it moves. Which type of agent is suitable?

A) Offline Agent
B) Online Search Agent
C) Static Agent
D) Deterministic Agent

Answer: B) Online Search Agent

11. Why is an admissible heuristic important?

A) It reduces memory usage
B) It ensures optimality
C) It increases branching factor
D) It avoids loops

Answer: B) It ensures optimality

12. What is the primary benefit of Alpha-Beta pruning?

A) Reduces depth of tree
B) Reduces number of nodes evaluated
C) Changes game rules
D) Improves heuristic accuracy

Answer: B) Reduces number of nodes evaluated

13. What is the main drawback of Hill Climbing?

A) High memory usage
B) Revisiting nodes
C) Getting stuck in local maxima
D) Requires complete tree

Answer: C) Getting stuck in local maxima

14. What ensures a valid CSP solution?

A) Heuristic function
B) Path cost
C) Constraint satisfaction
D) Depth-first traversal

Answer: C) Constraint satisfaction

15. What is the role of an evaluation function?

A) Generate successors
B) Estimate utility of a state
C) Reduce branching factor
D) Store moves

Answer: B) Estimate utility of a state

16. In A* search, a node has a path cost of 7 and a heuristic estimate of 5. What is the value of f(n)?

A) 12
B) 2
C) 35
D) 7

Answer: A) 12

17. A heuristic function overestimates the actual cost to reach the goal. What happens?

A) A* remains optimal
B) A* becomes non-optimal
C) A* becomes BFS
D) A* stops working

Answer: B) A* becomes non-optimal

18. For a search tree with branching factor 2 and depth 4, what is the approximate number of nodes in a full binary tree?

A) 10
B) 15
C) 31
D) 16

Answer: C) 31

19. In a minimax tree, each node has 3 children and the depth is 3. How many leaf nodes are present?

A) 9
B) 27
C) 6
D) 12

Answer: B) 27

20. What is the best-case time complexity of Alpha-Beta pruning?

A) O(b^d)
B) O(b^(d/2))
C) O(d^b)
D) O(log b)

Answer: B) O(b^(d/2))

Quick Answer Summary

1. C
2. C
3. B
4. B
5. B
6. B
7. B
8. B
9. C
10. B
11. B
12. B
13. C
14. C
15. B
16. A
17. B
18. C
19. B
20. B